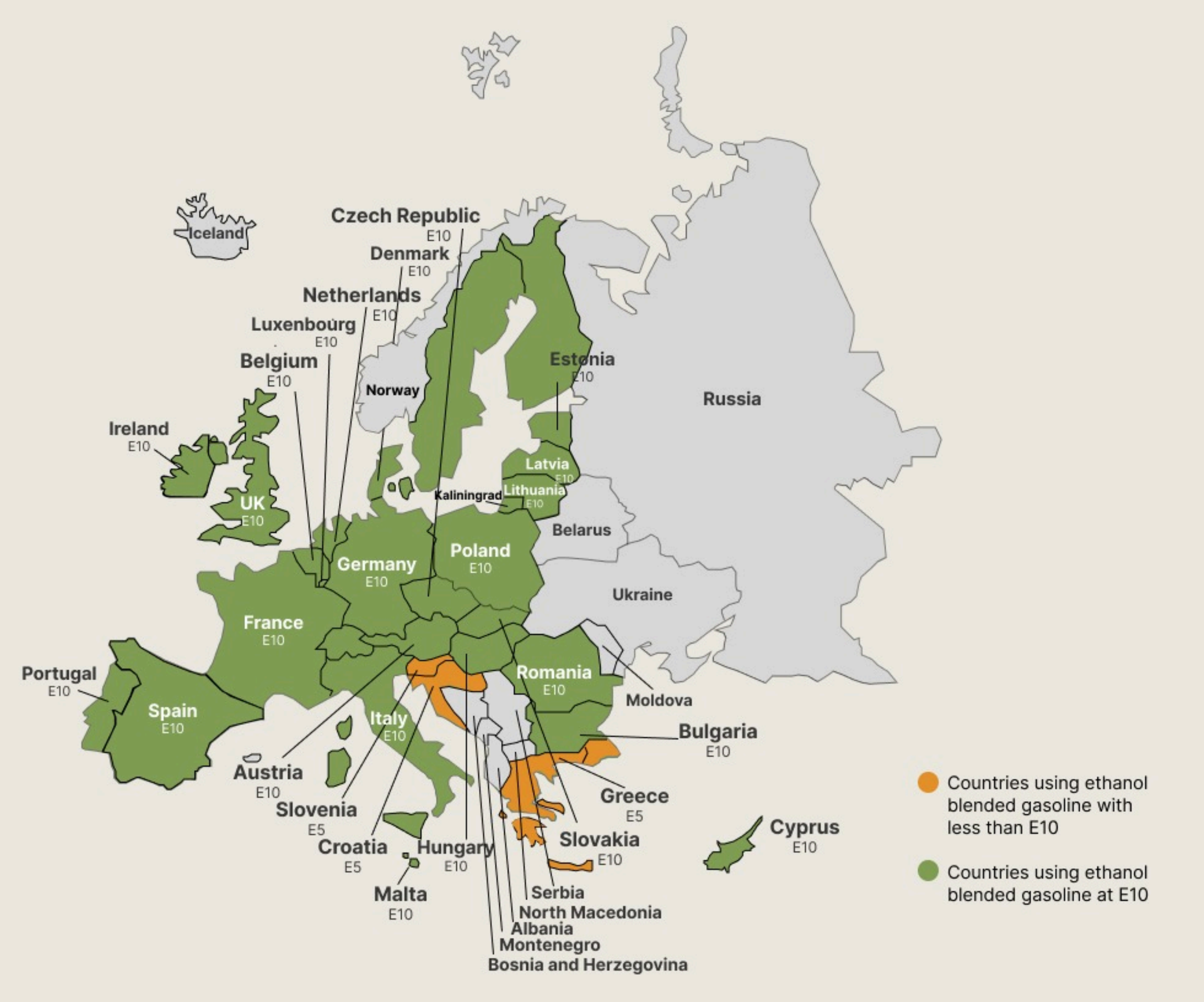
Ethanol Blending in Gasoline - Germany

September, 2025

Ethanol Blending in Europe

There are important fuel quality and environmental impact of vehicle emission challanges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

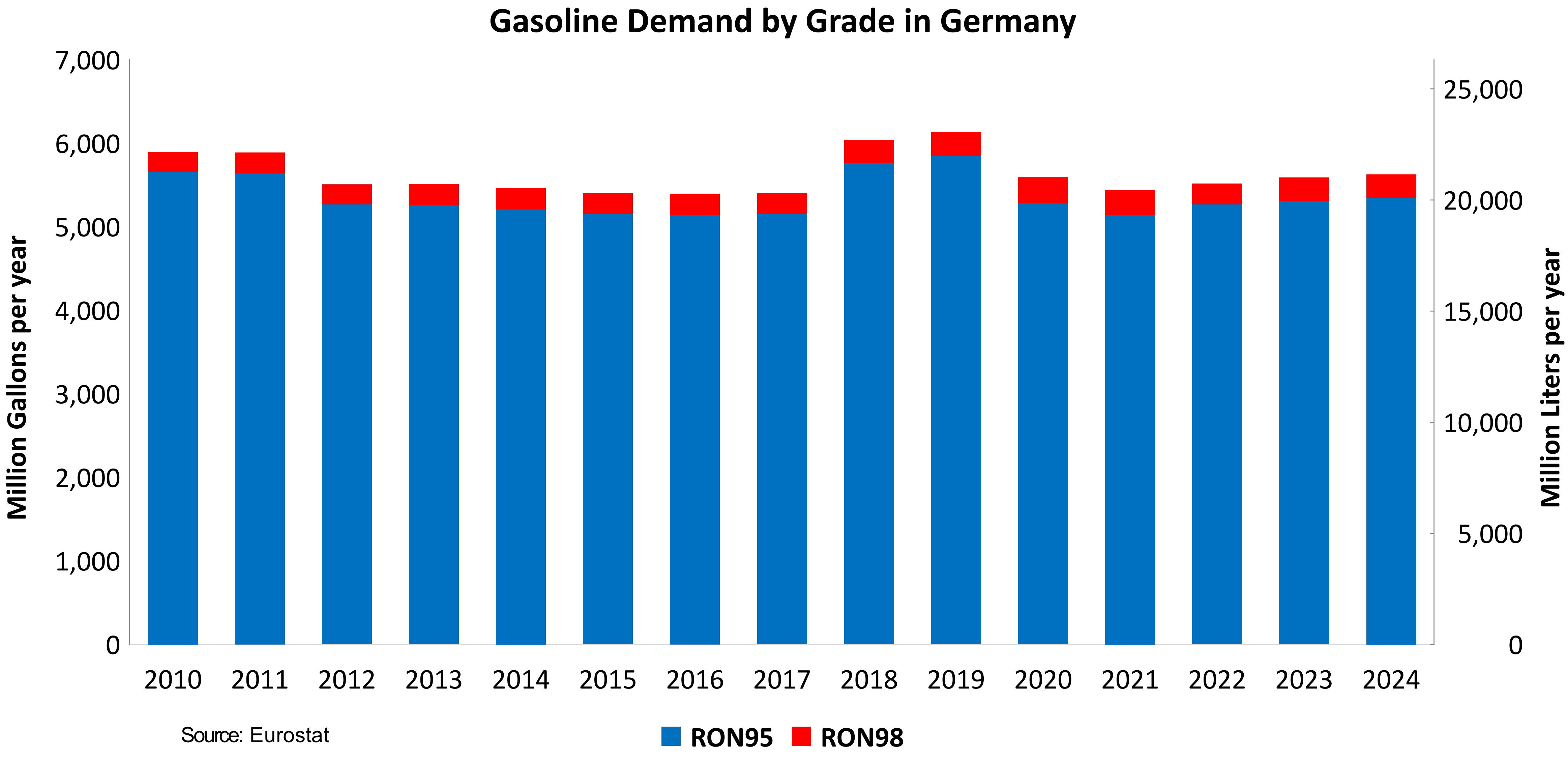
Ethanol Gasoline Blending in Germany

In 2024, gasoline consumption in Germany was 21,292 million liters (5,625 million gallons) and growth rate was -1.7%. Gasoline production and net imports were 22,284 and -3,611 million liters (5,887 and -954 million gallons) correspondingly. Germany complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.2 RON.

In 2024, ethanol consumption in Germany was 2,262 million liters (598 million gallons) representing 10.6% of gasoline demand. Ethanol production and Net Imports were 1,333 and 929 million liters (598 and 245 million gallons) correspondingly. Ethanol sources are Cereals 64% and Sugar Beet 33%.

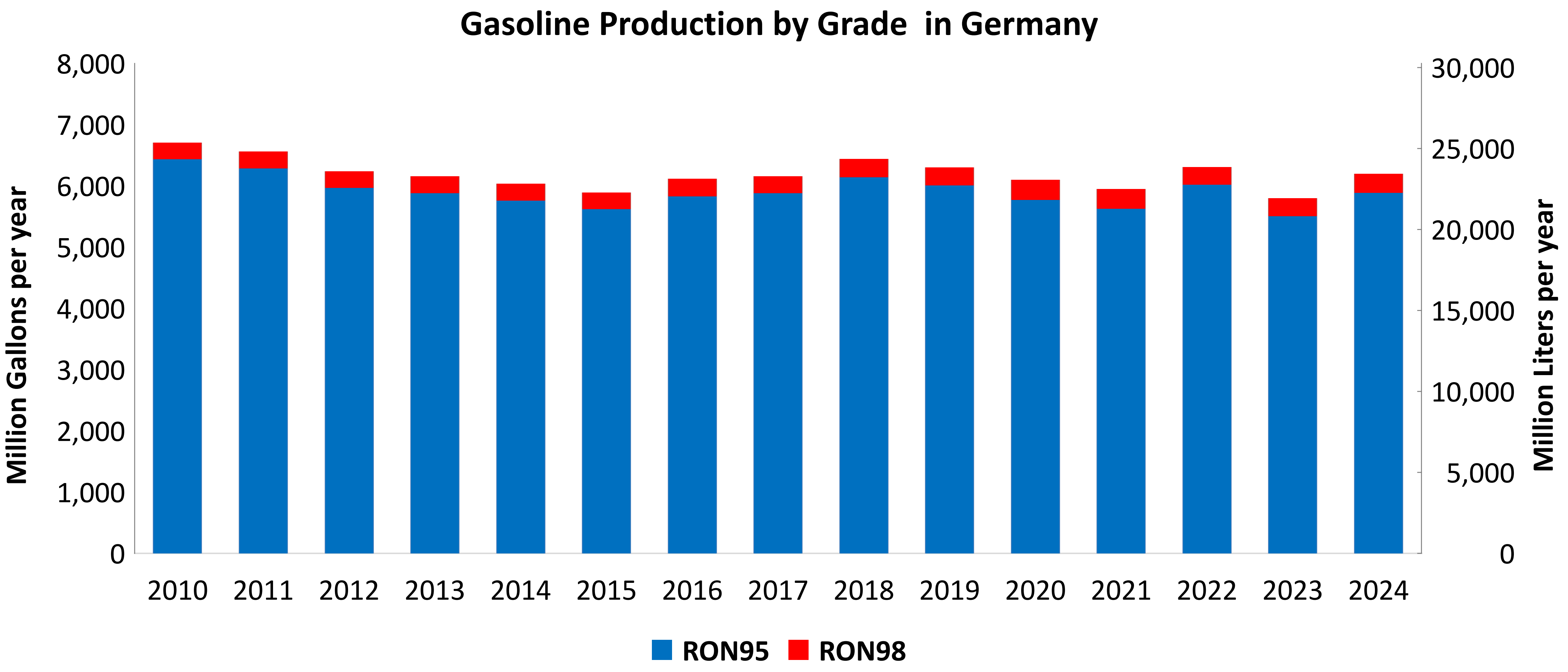
Source: Eurostat, European Commision

Gasoline Demand in Germany



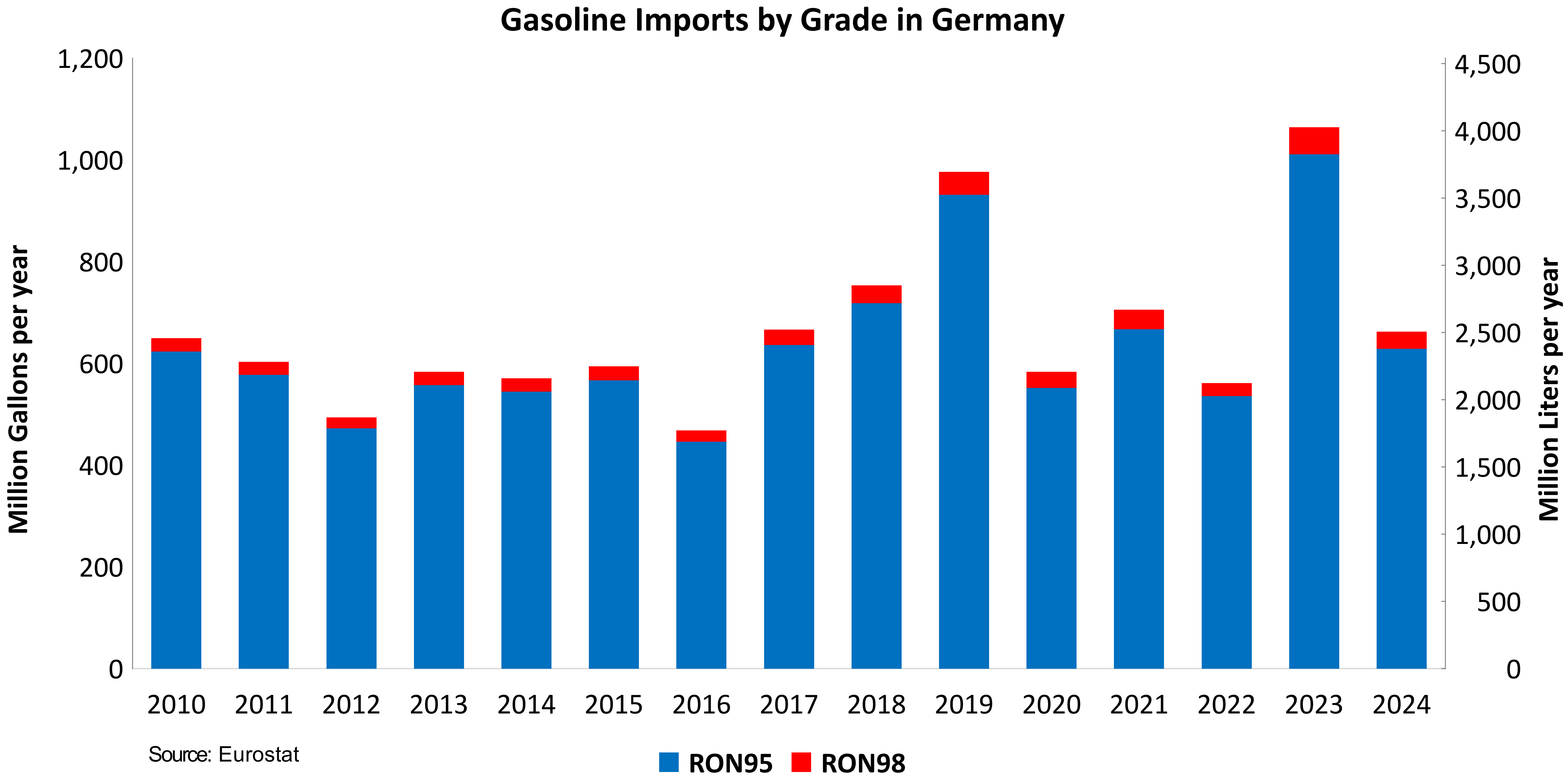
Source: Eurostat

Gasoline Production in Germany

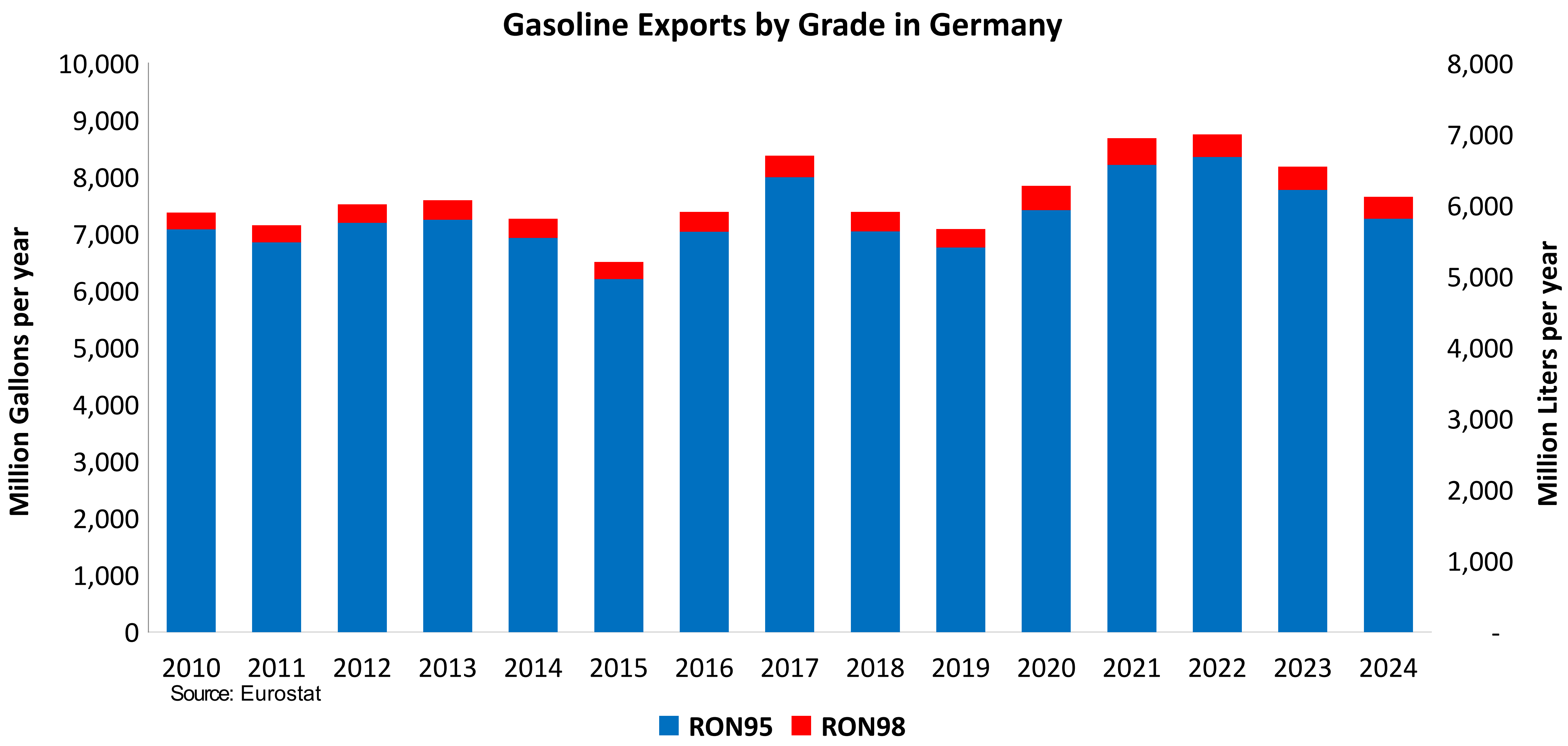


Source: Eurostat

Gasoline Imports in Germany

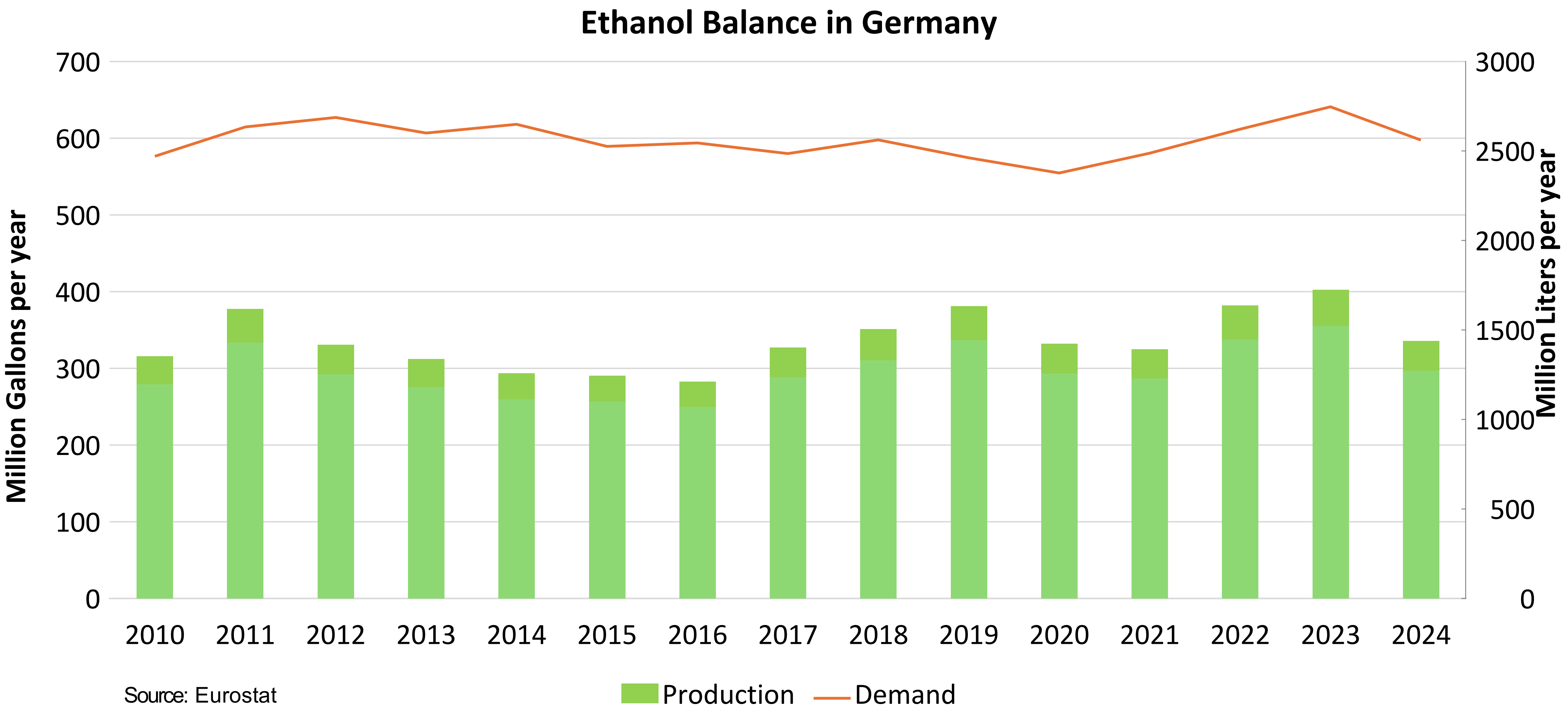


Gasoline Exports in Germany



Source: Eurostat

Ethanol Balance in Germany



Source: Eurostat

Gasoline Quality in Germany

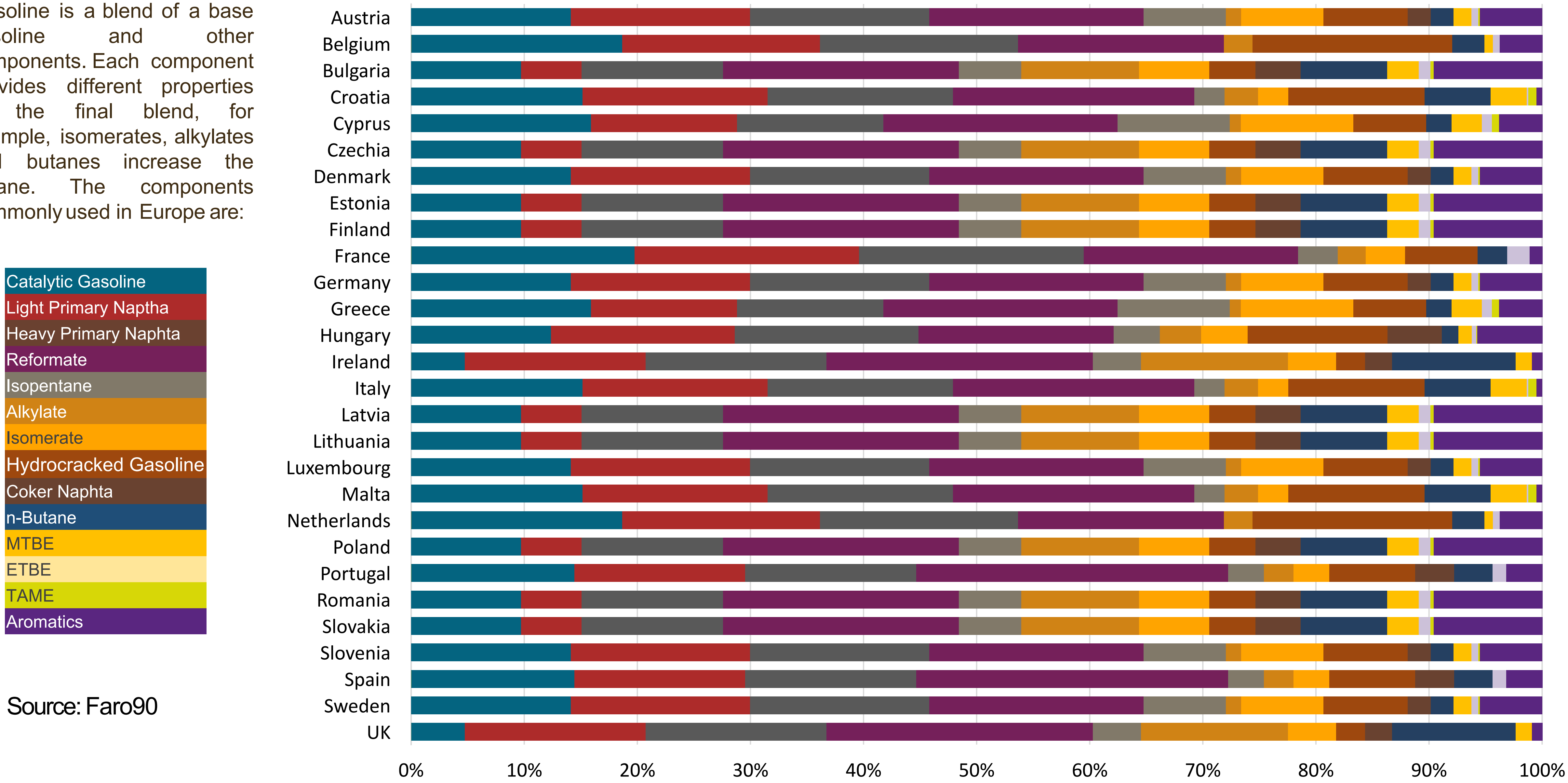
Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	0.7							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	10.5							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Gasoline Component Blending in Europe

Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:



Source: Faro90

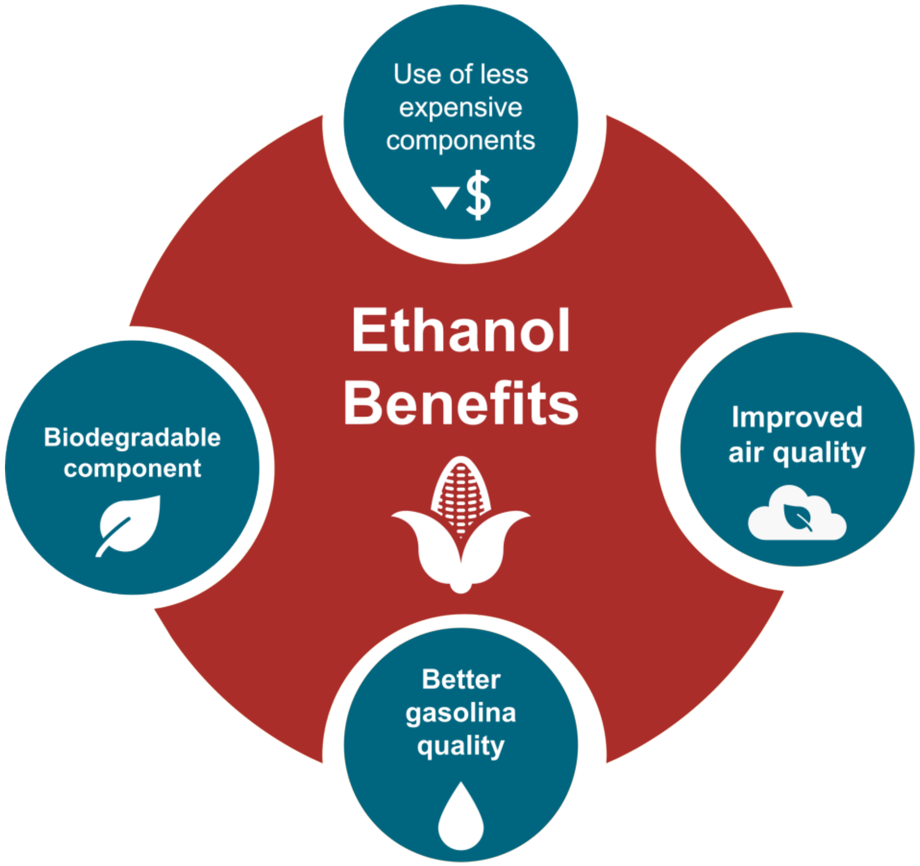
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component.The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives.To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

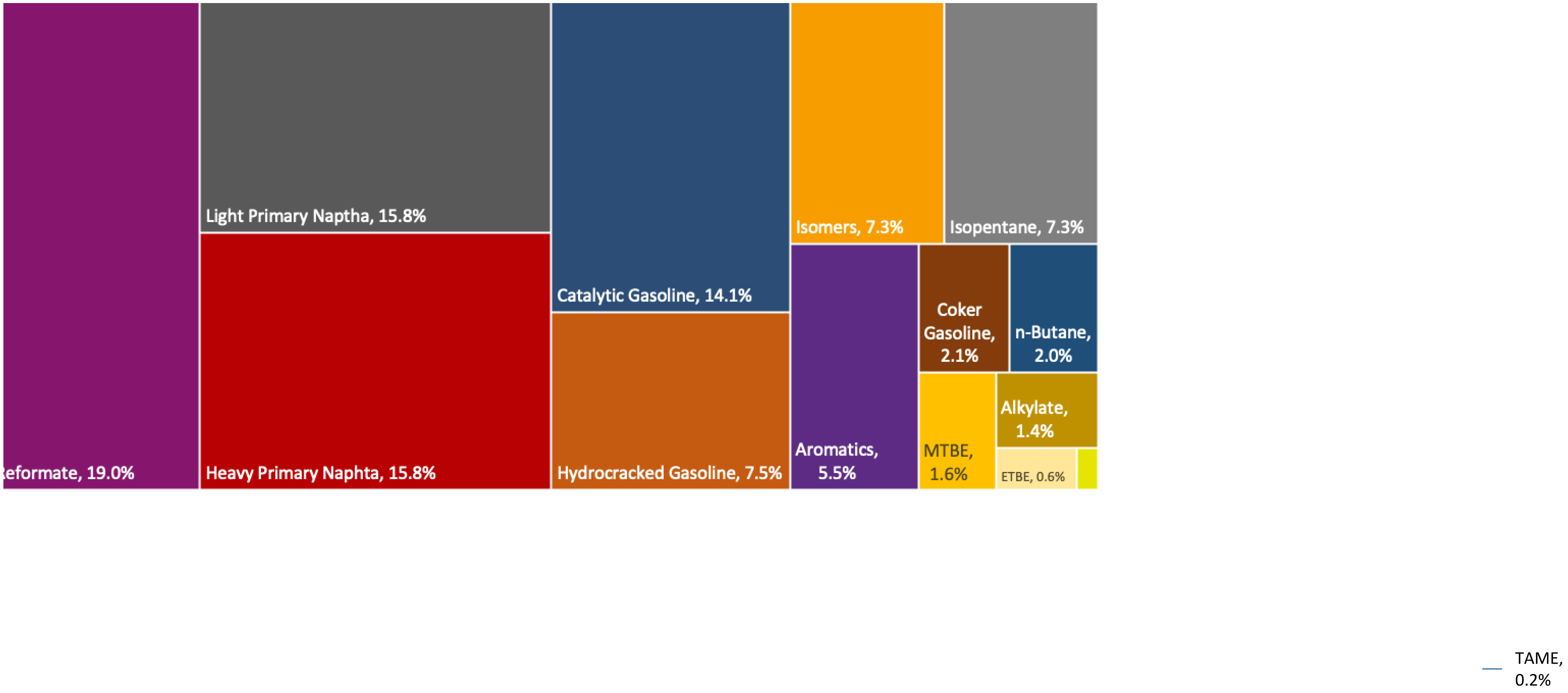
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

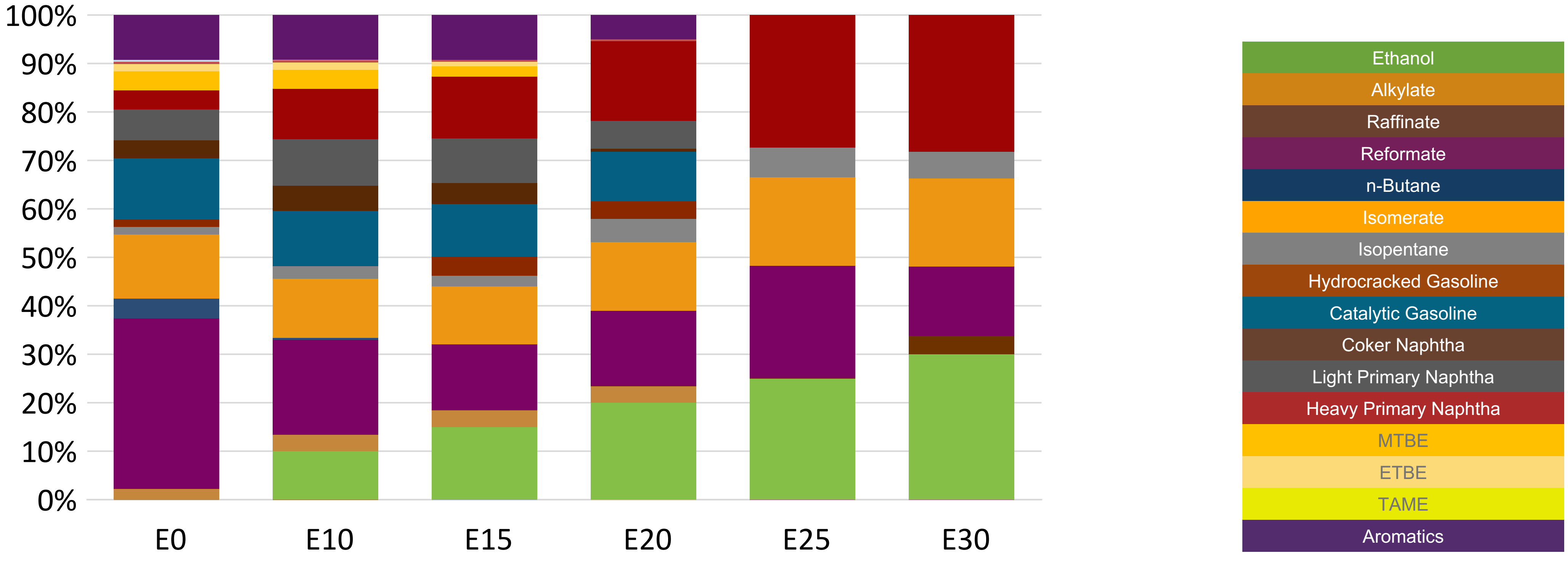


Germany Available Blending Components



Source: Faro90

Germany – Gasoline 95 – Constant Octane

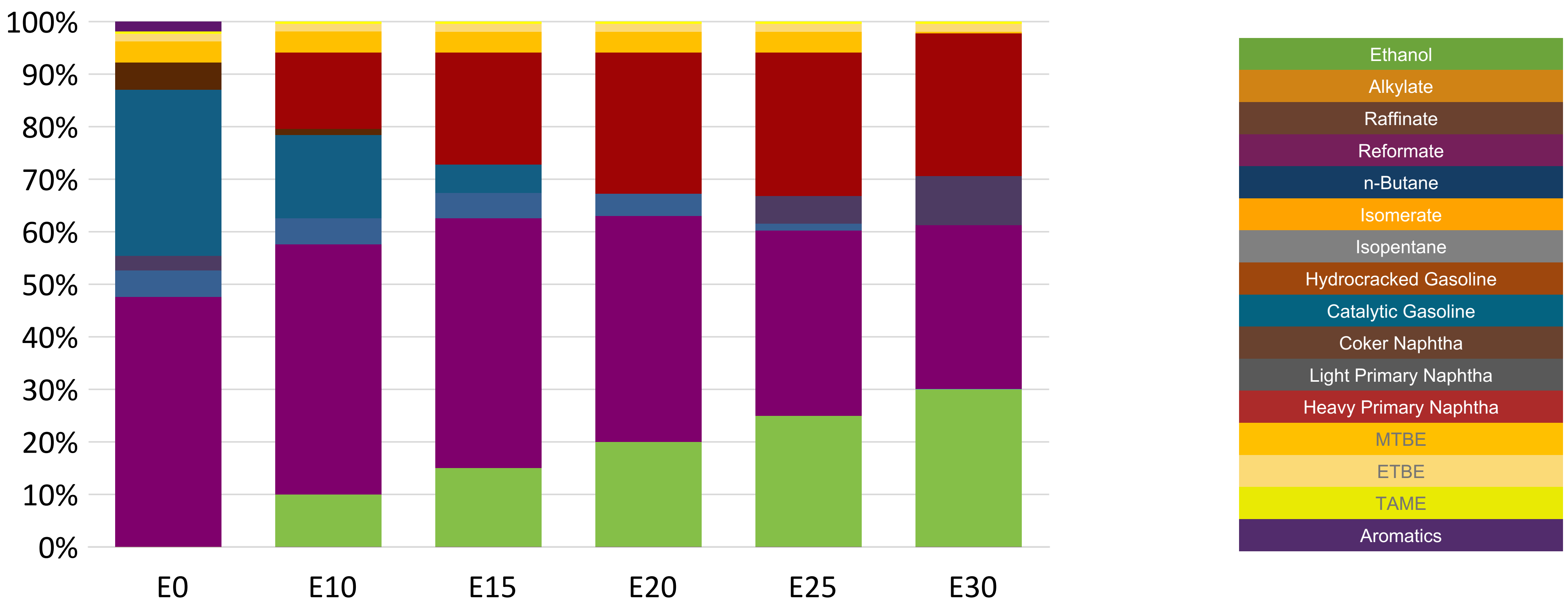


Octane (RON)	95.0	95.0	95.0	95.0	95.1	95.6
Price (US$/gal)	$ 2.272	$ 2.144	$ 2.083	$ 2.002	$ 1.880	$ 1.814

Average CIF 2024 Prices

Source: Faro90

Germany - Gasoline 98 - Constant Octane

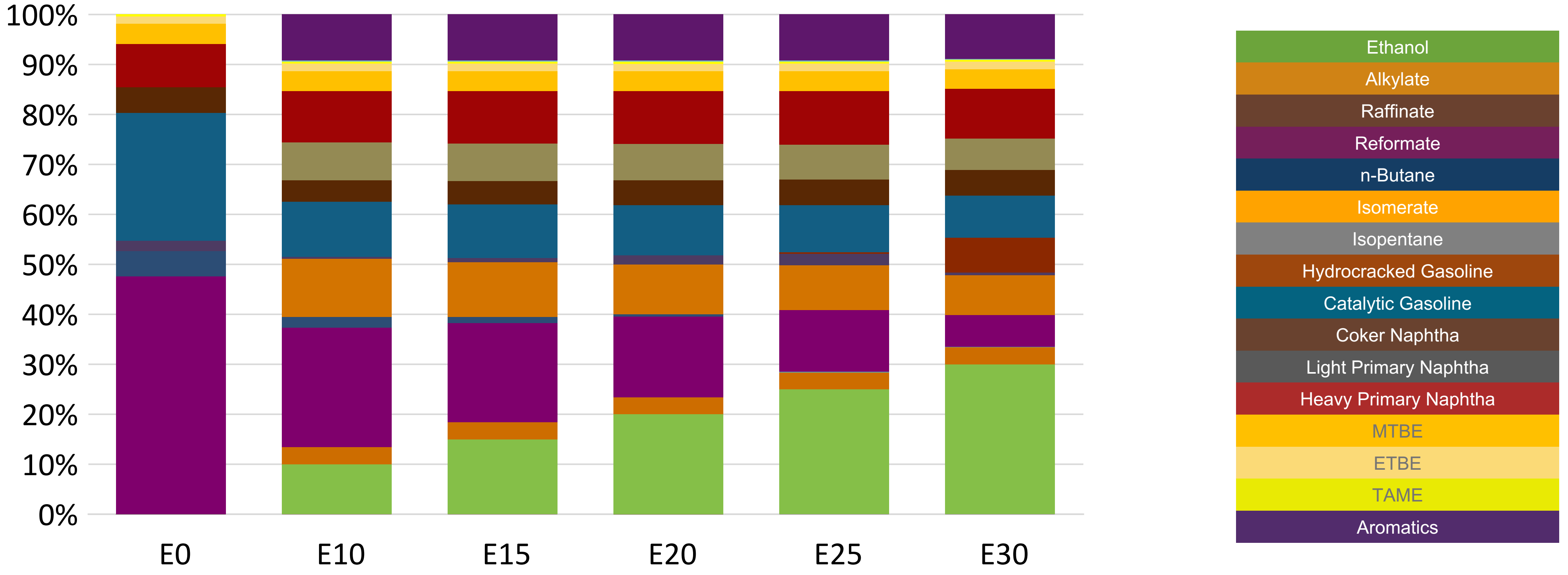


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.347	$ 2.218	$ 2.146	$ 2.069	$ 1.992	$ 1.925

Average CIF 2024 Prices

Source: Faro90

Germany – Gasoline 95 – Increased Octane

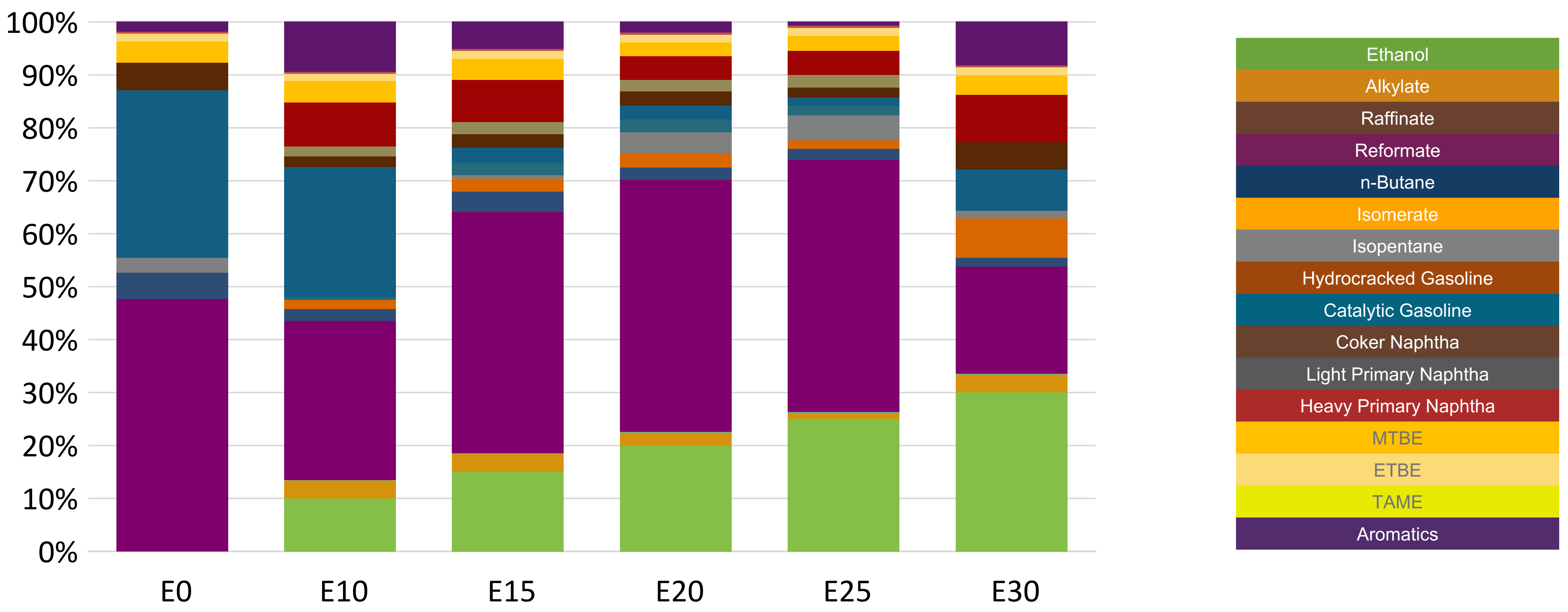


Octane (RON)	95.0	96.5	97.9	99.4	100.8	102.0
Price (US$/gal)	$ 2.209	$ 2.209	$ 2.209	$ 2.209	$ 2.209	$ 2.209

Average CIF 2024 Prices

Source: Faro90

Germany – Gasoline 98 – Increased Octane



Octane (RON)	98.0	99.2	101.3	102.6	104.0	104.7
Price (US$/gal)	$ 2.347	$ 2.347	$ 2.347	$ 2.347	$ 2.347	$ 2.347

Average CIF 2024 Prices

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

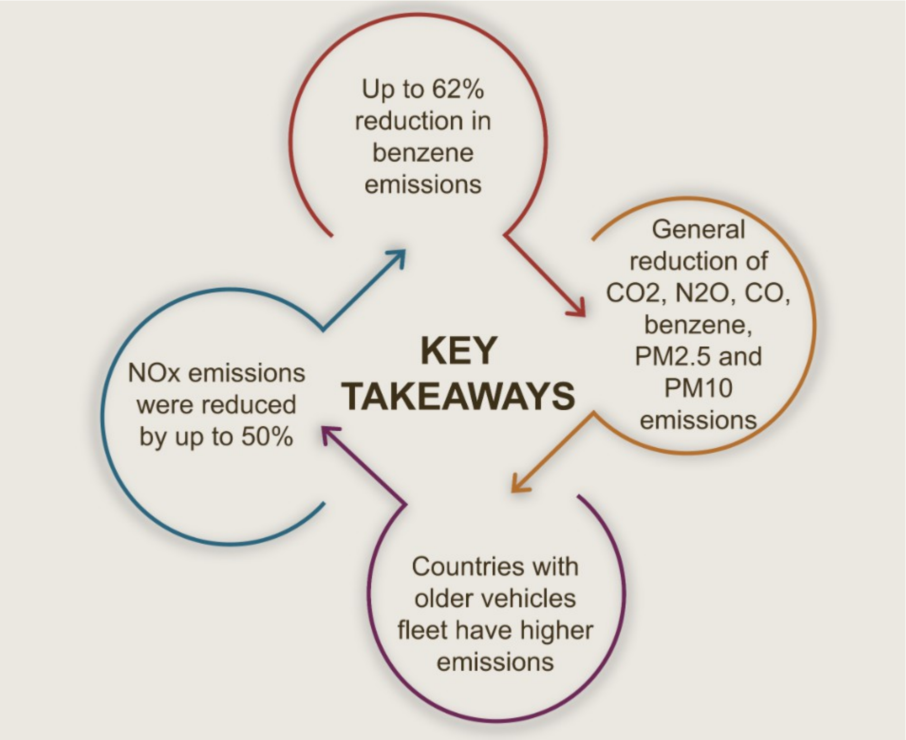
The model used in this analysis takes as a reference the **International Vehicle Emissions Model (IVE).**
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



Source: Bureau of transportation statistics.

Gasoline Vehicle Fleet - Germany

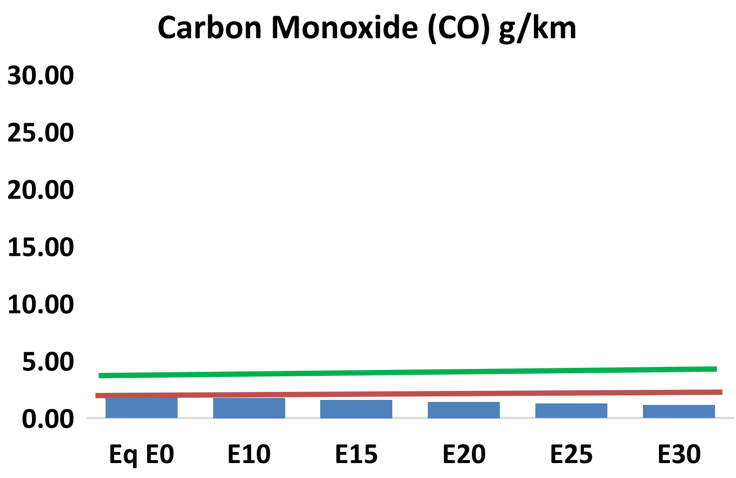
Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	19,353
		Single FI	3-Ways		>8 years	174,176
		Multi FI	3-Ways		>8 years	1,655,439
			Euro III		>8 years	1,603,283
			Euro IV		>8 years	5,474,789
			Euro V		>8 years	2,555,744
			Euro 6		>8 years	4,974,662
					4-8 years	10,980,947
					<4 years	16,791,954
			Hybrid		All	2,100,972
		Electric			All	814,320
	Medium	Carburator	3-Ways	PCV Tank	>8 years	1,163
		Single FI	3-Ways		>8 years	10,469
		Multi FI	3-Ways		>8 years	86,505
			Euro III		>8 years	21,876
			Euro IV		>8 years	36,990
			Euro V		>8 years	13,890
			Euro 6		>8 years	42,183
					4-8 years	67,255
					<4 years	106,528
			Hybrid		All	1,757
		Electric			All	2,010
Sml Engines	Light	4-Cycle	None	PCV	>8 years	1,256,474
			Euro III		>8 years	604,481
			Euro IV		>8 years	806,968
			Euro V		>8 years	263,309
			Euro 6		>8 years	482,272
					4-8 years	616,089
					<4 years	1,061,737
			Hybrid		All	1,010
		Electric			All	40,510
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	40,285
		Fuel-Injection	3-Ways		>8 years	3,198,889
			Euro III		>8 years	268,027
			Euro IV		>8 years	494,603
			Euro V		>8 years	207,082
			Euro 6		>8 years	828,974
					4-8 years	1,383,193
					<4 years	2,009,814
			Hybrid		All	788
		Electric			All	691

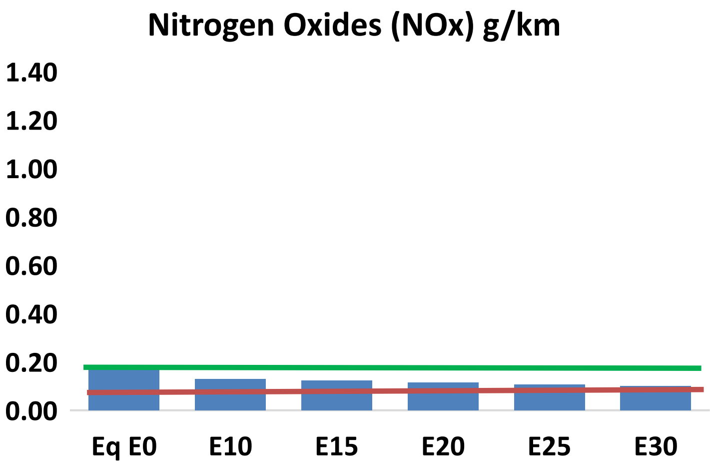
Vehicle Fleet: 61,101,463
Gasoline Vehicle Fleet: 32,331,793

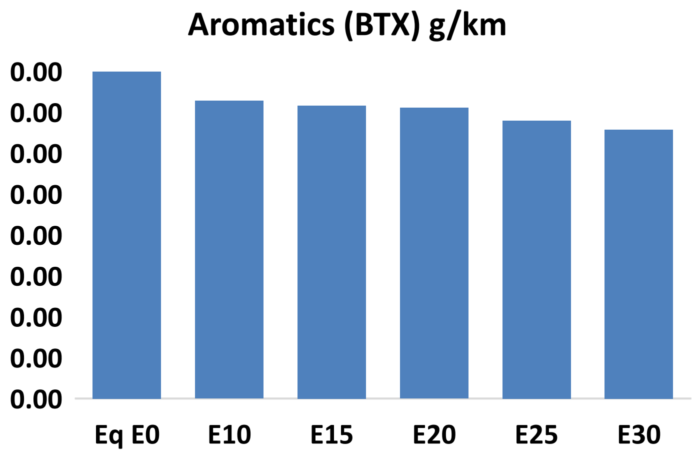
Source: Eurostat, ACEA

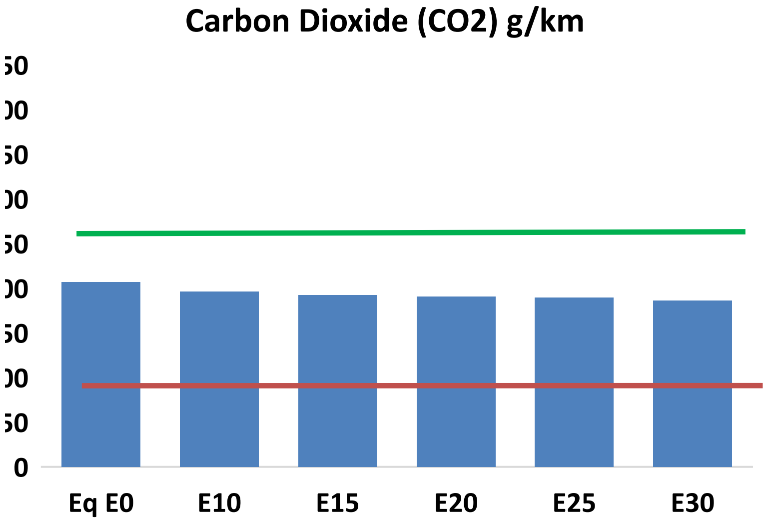
Germany – Vehicular Emissions

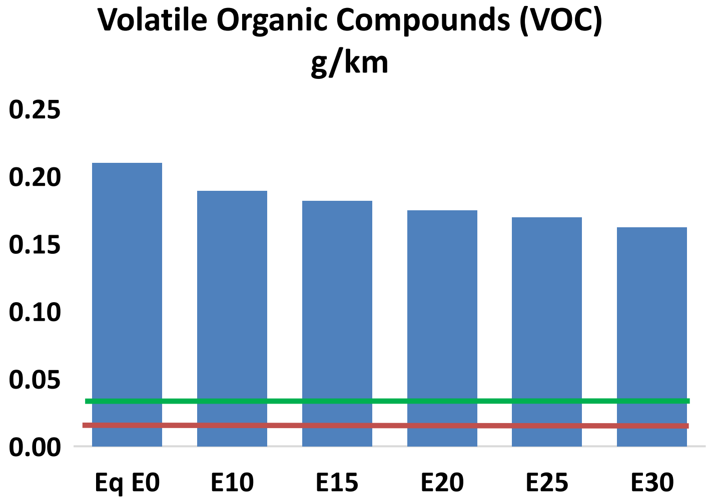
TIER III BIN 125 United States
Euro 6 Standard

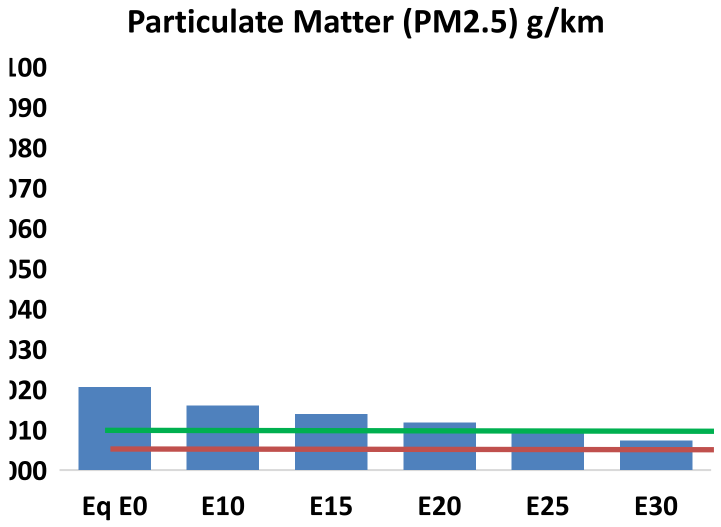












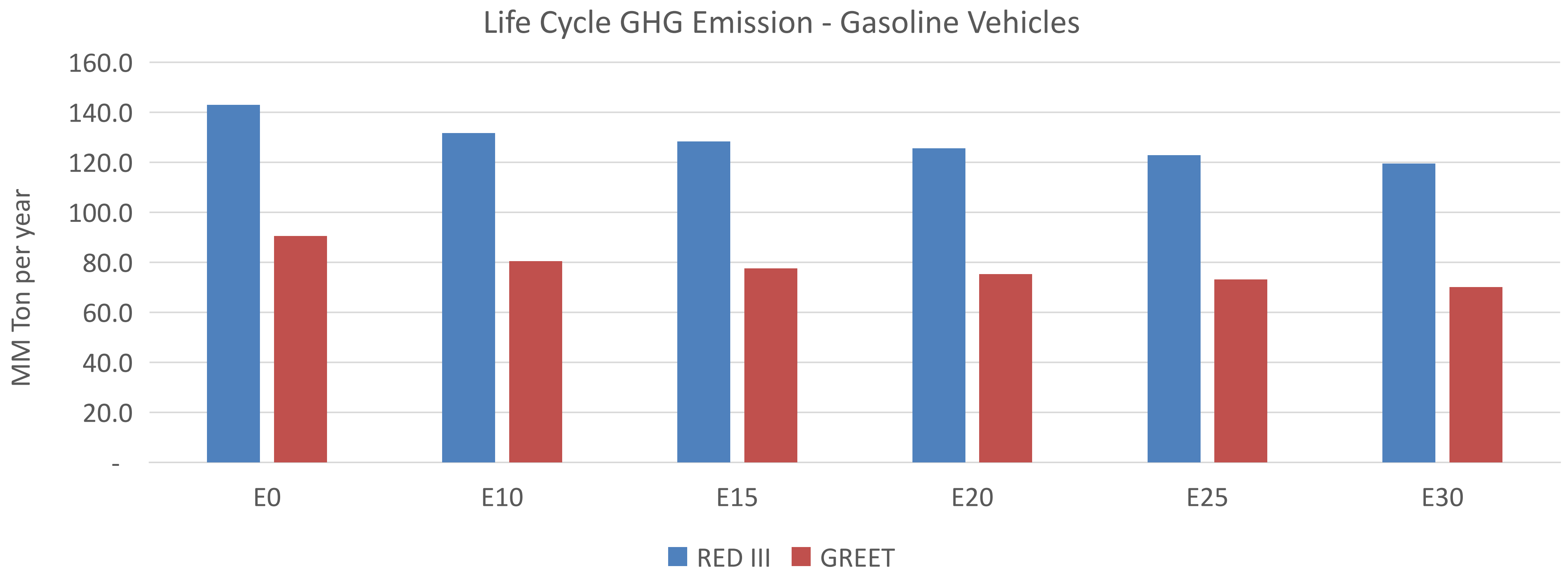
Source: Faro90

Germany – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	1,772,395	1,605,976	1,439,558	1,297,299	1,151,430	1,038,448	921,561	-19%	-35%
CO2	169,687,313	165,323,925	161,202,948	157,961,920	156,358,133	155,431,613	152,566,562	-5%	-8%
VOC	172,340	163,809	155,277	149,207	143,521	139,277	133,163	-10%	-17%
NOx	152,188	114,141	106,531	100,406	94,691	88,391	81,725	-30%	-38%
SOx	1,977	1,827	1,678	1,551	1,430	1,299	1,161	-15%	-28%
NH3	58,379	57,801	57,223	57,495	58,141	58,596	58,974	-2%	0%
N2O	1,820	1,810	1,803	1,811	1,849	1,869	1,891	-1%	2%
CH4	41,275	41,275	41,275	41,894	42,802	43,586	44,329	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	1,662	1,583	1,505	1,451	1,401	1,363	1,309	-9%	-16%
Acetaldehyde	3,372	3,870	5,679	8,648	11,783	13,464	15,909	68%	249%
Formaldehyde	10,885	10,583	12,337	14,486	15,176	16,789	18,277	13%	39%
Benzene	3,279	3,151	2,986	2,939	2,914	2,787	2,697	-9%	-11%
PM2.5	10,785	9,576	8,368	7,267	6,171	5,007	3,795	-22%	-43%
PM10	6,036	5,360	4,683	4,067	3,453	2,802	2,124	-22%	-43%

Source: Faro90

Germany – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	717.1
Target @ 2035 (MMT/year)	403.6
Delta	313.4

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	11.2%	14.4%	16.9%	19.3%	22.6%
RED II/III	0.0%	7.9%	10.3%	12.2%	14.1%	16.4%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	3.2%	4.2%	4.9%	5.6%	6.5%
RED II/III	0.0%	3.6%	4.7%	5.6%	6.4%	7.5%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90